10.6 Comparing Data Sets

Lesson Introduction

Benchmark	MA.7.DP.1.2 Given two numerical or graphical representations of data, use the measure(s) of center and measure(s) of variability to make comparisons, interpret results, and draw conclusions about the two populations.	Learning Target	• I can compare and interpret two numerical representations of data for two populations.

	Building On	Working Toward
Benchmark Coherence	MA.6.DP.1.3 Given a box plot within a real-world context, determine the minimum, the lower quartile, the median, the upper quartile, and the maximum. Use this summary of the data to describe the spread and distribution of the data. MA.6.DP.1.4 Given a histogram or line plot within a real-world context, qualitatively describe and interpret the spread and distribution of the data, including any symmetry, skewness, gaps, clusters, outliers, and the range.	MA.912.DP.2.1 For two or more sets of numerical univariate data, calculate and compare the appropriate measures of center and measures of variability, accounting for possible effects of outliers. Interpret any notable features of the shape of the data distribution.

Essential Questions	• How can measures of center and variation be used to compare data sets? • Why do you use different measures of center and variation to describe different data sets? • How do outliers impact the comparison of data sets?
Lesson Narrative	This lesson builds on the previous one to continue comparing data sets using measures of center and variation. Students begin by working collaboratively to determine which measures of center and variation should be used to describe data sets and then apply those measures to determine which methods are best suited for various scenarios based on their data presented.

Component Summary	10.6.1 Warm-Up (~ 5 min.)	Observe line plots to record what students notice and wonder *(SE p. 134)*	10.6.4 Activity (10-15 min.)	Apply measures of center and variation to describe and compare two data sets *(SE p. 136)*
	10.6.2 Exploration (10-15 min.)	Determine measures of center and variation to compare two data sets in context *(SE p. 135)*	10.6.5 Your Turn! (5-10 min.)	Analyze two data sets to determine the most effective means of communication *(SE p. 137)*
	10.6.3 Try It (5-10 min.)	Compare data sets to determine the best way to get to school *(SE p. 136)*	10.6.6 Wrap-Up (~5 min.)	Students explain why they do or do not agree with a conclusion based on the data in the previous component *(SE p. 138)*

	Glossary	Materials	Additional Preparation
Resources	Key Terms: N/A Additional Terms: N/A	• N/A	• N/A

<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>Students may struggle to explain their thinking when comparing data sets.
Students may struggle to understand which method of communication is most effective in 10.6.5 Your Turn! activity.</td><td>Consider providing sentence stems for additional scaffolding for written response questions.
Allowing students to use calculators during the lesson will help students focus on comparing and interpreting the data. Consider having no-calculator questions where students turn their calculator over.
If time is an issue, then consider assigning 10.6.5 Your Turn! for homework.</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Think-Pair-Share
10.6.3 Try It creates multiple opportunities for students to engage in Think-Pair-Share routines. As students work with their partners, formalize this routine to allow them to formalize their responses before putting them in writing.</td><td>MA.K12.MTR.4.1: Discussion and Reflection
Partner discussion is embedded throughout this lesson, as almost every component is collaborative. Students are given the opportunity to communicate their mathematical ideas and methodology through discourse, particularly in the 10.6.4 Activity: Guess My Data Sets. Throughout this activity students justify their own reasoning and compare that to the reasoning of their peers.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">Notice 10.6.4 Activity provides a variety of entry points for learners to engage with the content. This activity is partner-based, offering space for students to verbally reason with a partner and work independently. Some students may prefer to complete problems before discussing with a partner, while others may benefit from verbally processing their reasoning before finalizing their response. All students will benefit from the mathematical discourse.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
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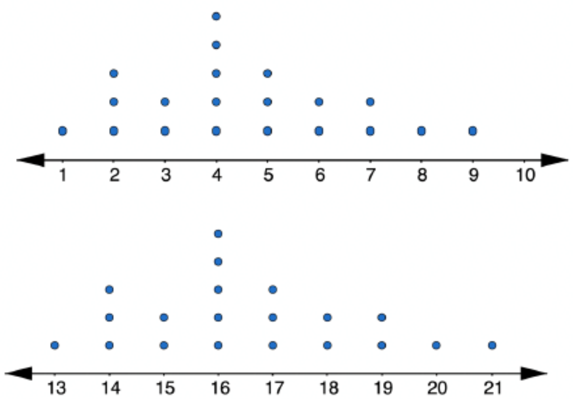
10.6.1 Warm-Up: Notice and Wonder

Component Overview: *Students consider two line plots and record what they notice and wonder about what the displays show.*

10.6.1 Warm-Up: Notice and Wonder

The line plots represent the distribution of the tip amounts, in dollars, left at two different restaurants on the same night.



Tip Amount (in dollars)

1. Complete each statement.

   a. When looking at the data displays, one thing I notice is ...
   Answers will vary. The shape of the distributions is the same, but they are shown over different scales.

   b. One thing I wonder is ...
   Answers will vary. Why do workers at one restaurant receive higher tips than the other?

Notice and Wonder
Position students to continue to conceptually understand measures of center and variation for two data sets. Engage students in a Notice and Wonder routine. Ask questions like:

- *What do you notice about the data sets?*
- *What do the line plots have in common?*
- *What is different?*
- *What does this make you wonder?*

10.6.2 Exploration: Calculating Measures of Center and Variation

Component Overview: *Students work with a partner to find the measures of center and variation for two data sets and then analyze the data to draw conclusions about each set. After completing statements describing the data, students explain why the measures of variation and center chosen to summarize the data sets were chosen and check their work with their partner.*

10.6.2 Exploration: Calculating Measures of Center and Variation

On a college entrance exam, the possible scores on the math section range from 200-800. Below are the scores for 20 randomly selected female high school seniors.

454 409 437 415 452 440 454 478 401 403 426 426 437
465 465 429 470 448 511 453

The scores for 20 randomly selected male high school seniors are shown below.

408 440 414 391 480 415 426 392 469 446 409 432 375
388 413 451 370 415 487 404

1. Work with your partner to complete the table by calculating the measures of center and variability for each set of data.

Measure	Female Students	Male Students
Mean	443.65	421.25
Median	444	414.5
Range	110	117
IQR	33.5	45
Potential Outliers	511	*none*

2. Complete the statements.

   a. Female students tend to have ○ lower ○ higher scores on the math section of the college entrance exam, with a median of 444, than males, who have a median of 414.5.

This Exploration is scaffolded to allow students to engage with it with their partner rather than being guided through by the teacher. There will be an opportunity to summarize and synthesize what is discovered in the next component. Students may revise their thinking throughout the Exploration as they engage with each new Reflection.

Observe and listen as students work with their partners. Make note of student ideas that you can ask them to share in the whole group at the conclusion of the component.

10.6.2 Exploration: Calculating Measures of Center and Variation (continued)

b. Scores for males tend to have
- less
- more

variability, with an IQR of 45, than scores for females, with an IQR of 33.5.

c. Explain why the measures median and IQR were chosen to describe the data sets above, instead of mean and range.
Answers will vary. The mean and range are not accurate measures of center and variation for comparison because 511 is an outlier for the female students' data.

d. Check your work on Parts A-C with your partner and if necessary, resolve any differences.

Additional scaffolding can be provided through the use of questioning for students struggling to explain:
- *Did outlier(s) exist in the data?*
- *Do outliers play a role in your deciding which measures to use?*

10.6.3 Try It: Drawing Conclusions

__Component Overview:__ Students work with a partner to compare the measures of center and spread for two data sets to determine if a student should ride his bike to school or take a car. Students make arguments for both sides based on the data in the tables.

10.6.3 Try It: Drawing Conclusions

Shawn wants to determine the best way to get to school. Over several weeks, he took turns riding in a car and riding his bicycle and recorded the time, in minutes.

Car	
19	17
15	18
12	22
29	14

Bicycle	
24	25
27	26
21	28
32	23

Shawn calculated the measures of center and spread for each transportation method. They are shown in the table.

	Car	Bicycle
Mean	18.25	25.75
Median	17.5	25.5
Range	17.0	11.0
IQR	6.0	4.0

1. Make an argument as to why Shawn should ride in a car to get to school.
   Answers will vary. The average time it takes to get to school by car is less than the average time it takes to travel by bicycle. There is also a larger range in the data, so the times may change, but overall it's faster regardless. Plus, he can leave later in the morning.

2. Make an argument as to why Shawn should ride his bicycle to school.
   Answers will vary. Shawn should ride his bicycle to school because the time it takes him to get to school on his bike varies less than the time it takes to travel by car. Even though the car is faster, he can better predict how long it will take to get there on his bike when deciding what time to leave in the morning.

Think-Pair-Share
Engage students in a Think-Pair-Share routine before they complete each written response question. This will give students the opportunity to formalize their responses through discussion before putting them in writing, as well as to receive and provide feedback to partners.

Additional scaffolding can be provided through the use of sentence stems:

- *Shawn should take a car/bike to school because ….*
- *I know this because the measure of center/variation shows ….*

10.6.4 Activity: Guess My Data Sets

Component Overview: *Students work with a partner to play "Guess My Data Sets". Each partner chooses 2 data sets and writes a paragraph comparing them using measures of center and variability. Both partners read their descriptions while the other partner guesses which two data sets are being described and then records their answers in a table.*

10.6.4 Activity: Guess My Data Sets

Suppose an English teacher takes five random samples of students and records their most recent test scores.

Data Set A: $\{22, 0, 44, 40, 56, 32, 64, 44, 52, 60\}$

Data Set B: $\{60, 62, 62, 60, 64, 68, 64, 62, 68, 60\}$

Data Set C: $\{70, 82, 74, 76, 80, 72, 84, 70, 78, 82\}$

Data Set D: $\{72, 86, 100, 76, 92, 90, 100, 78, 98, 86\}$

Data Set E: $\{64, 36, 74, 82, 96, 80, 66, 78, 68, 0\}$

Without telling your partner, choose two data sets from above.

1. Write a paragraph comparing the two data sets you have chosen using measures of center and variability. Make sure to refer to the data sets as "Data Set 1" and "Data Set 2" in the paragraph you write.

Data Set	Mean	Median	Range	IQR	Outliers?
A	41.4	44	64	24	0
B	63	62	8	4	None
C	76.8	77	14	10	None
D	87.8	88	28	20	None
E	64.4	71	96	16	0 or 36

Answers will vary based on data sets chosen. Data Set 1 (C) has a mean of 76.8 where Data Set 2 (D) has a mean of 87.8. The median for Data Set 1 is 77 and for Data Set 2 the median is 88. The IQR for both data sets is 10. Neither data set have any outliers.

Take Turns
Students take turns completing the work in this activity, first explaining the data sets they choose and then choosing which data sets match their partner's description.

Encourage students to ask any clarifying questions as they work through this component together and justify their descriptions if necessary. As students work, circulate and consider having students summarize their partners' descriptions.

10.6.4 Activity: Guess My Data Sets (continued)

2. Determine with your partner who is partner A and who is partner B. Write your names on the lines for each partner in the table.
   Answers will vary.
3. Partner A should read their paragraph from Number 1 to partner B, and partner B will guess which two data sets were used. Then, partner B will explain their reasoning for their guess. Both partners will summarize the guess and reasoning in the first row of the table.
   Answers will vary.
4. Then, switch roles so that partner B reads their paragraph while partner A makes a guess and explains their reasoning.
   Answers will vary.

Partner	Guess	Reasoning for Guess
Partner A: ____________		
Partner B: ____________		

5. Reveal to your partner which data sets were used and discuss any errors that may have led to incorrect conclusions.
   Answers will vary.

If time allows, challenge students to compare a third pairing of data sets together.

10.6.5 Your Turn!

Component Overview: *Students work independently to analyze text and email response times to determine which method of communication is the most effective. If time is an issue, then consider assigning Your Turn! for homework.*

10.6.5 Your Turn!

1. A company is trying to determine which method of communication is best to tell their employees important information. They send text messages to nine employees and emails to another nine employees. They record how many minutes it takes for the employees to reply.

Text	5	20	45	9	13	12	4	1	11
Email	25	15	35	24	30	19	29	38	12

a. Which measure of center best summarizes the data sets?

Text: ○ mean ○ median   Email: ○ mean ○ median

b. Which measure of variability best summarizes the data sets?

Text: ○ range ○ IQR   Email: ○ range ○ IQR

c. Calculate the measures of center and variability.

	Text	Email
Mean	13.3	25.2
Median	11	25
Range	44	26
IQR	12	15.5

d. Use the most appropriate measures of center and variability to compare the effectiveness of the two methods of communicating with employees.
Answers will vary. Analyzing the data sets measure of center and variability, email appears to be the more effective way to communicate with employees. The mean and median are close together, and there are no outliers in the data set.

Challenge students to make predictions about the efficacy of the two means of communication before finding the measures of center and variability. Have students predict which method will be more effective, then check their work after completing the table.

Students may say that texting is the more effective method of communication because the mean response time is lower than the mean response time for emails, discounting the large variability in text response times. Address this misconception through questioning:

- *Is a method more effective if the data is fairly consistent, or if there is more variability? Why?*
- *Which method is more consistent? How do you know?*

10.6.6 Wrap-Up: Cool Down

Component Overview: *Students use the data from the previous component to explain whether they agree that email is the best means of communication for the company to use.*

10.6.6 Wrap-Up: Cool Down

1. Use the data from Your Turn to complete the prompt below.

   Lucas recommends that the company emails its employees because of the smaller range in the data set. Explain why you agree or disagree with Lucas.
   Answers will vary. Lucas is correct about the range being smaller.

Consider using this Cool Down as a formative data point to inform small groups or partner pairings in the next lesson, as students continue to compare data sets using measures of center and variation.